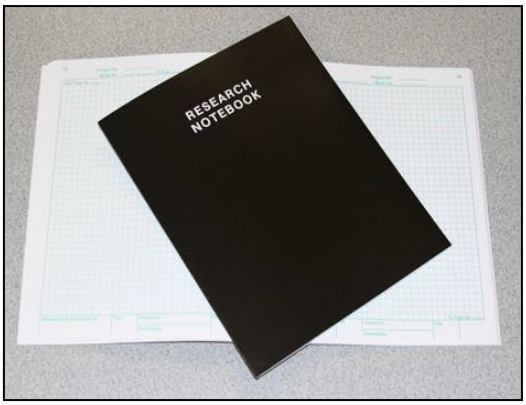
EXHIBIT 1

CEMM Paper Research Notebook Quick Tips[1]

- **Research notebooks should be used to capture *daily research activities* (ideally for a single project).**
- **Notebooks must cite the associated Quality Assurance Project Plan (QAPP) – Recommend using front page.**
- **Multiple users should be identified by a table with printed names and corresponding signatures/initials in the front of the notebook.**



- **Make** entries in **permanent ink** (**blue** or **black**) and **never** remove pages.
- **Document** each daily research activity with the following information:

  - Title/Topic of Activity (What?) - Results/Observations
  - Purpose of Activity (Why?) - Data Analysis
  - Materials and Methods (How?) - Conclusions/Next Steps

- **Sign (or initial) and date** each page of documented activity using the "recorded by" field.
- **Cross reference** documents/records (electronic files, SOPs, etc.), where appropriate.
- **Record** descriptive titles and page numbers in **Table of Contents**.
- **Affix** loose sheets with clear archival quality tape or acid-free glue. **Draw** a line across the corner of the affixed material and initial/date.
- **Record** information/data as it is taken or indicate insertion was made at a later date.
- **Never** leave blank areas. **Draw** a diagonal line through blank areas/pages and initial/date.
- **Strike through** errors with single line and initial/date. Include explanation for correction unless obvious transcription error. ~~Eaxmple~~ *AZ 8-17-2016* Example
- **Define** all abbreviations/acronyms in the front of the notebook or point to reference with info.
- **Include standard/reagent** prep info (e.g., date prepared, analyst, concentration, prep procedures, expiration date, material manufacturer/lot number/purity/conc./exp. date).

[1] ORD Policies and Procedures Manual Chapter 13, Section 13.2 "Scientific Recordkeeping: Paper" available at: https://webx.ord.epa.gov/sites/default/files/media/13.02_scientific_recordkeeping_paper-approved.pdf